

EXERCISE SHEET 12

GROUP AND RING THEORY

MAIN QUESTIONS - TO BE DONE BEFORE THE SEMINAR

12.1. Obtain the Smith normal form and rank for the following matrices over the given principal ideal domain R :

(a) $\begin{pmatrix} 0 & 2 & -1 \\ -3 & 8 & 3 \\ 2 & -4 & -1 \end{pmatrix}$, where $R = \mathbb{Z}$;

(b) $\begin{pmatrix} -x-3 & 2 & 0 \\ 1 & -x & 1 \\ 1 & -3 & -x-2 \end{pmatrix}$, where $R = \mathbb{Q}[x]$.

12.2. Find the invariant factors of the following matrix over $\mathbb{Q}[x]$:

$$\begin{pmatrix} 5-x & 1 & -2 & 4 \\ 0 & 5-x & 2 & 2 \\ 0 & 0 & 5-x & 3 \\ 0 & 0 & 0 & 4 \end{pmatrix}$$